

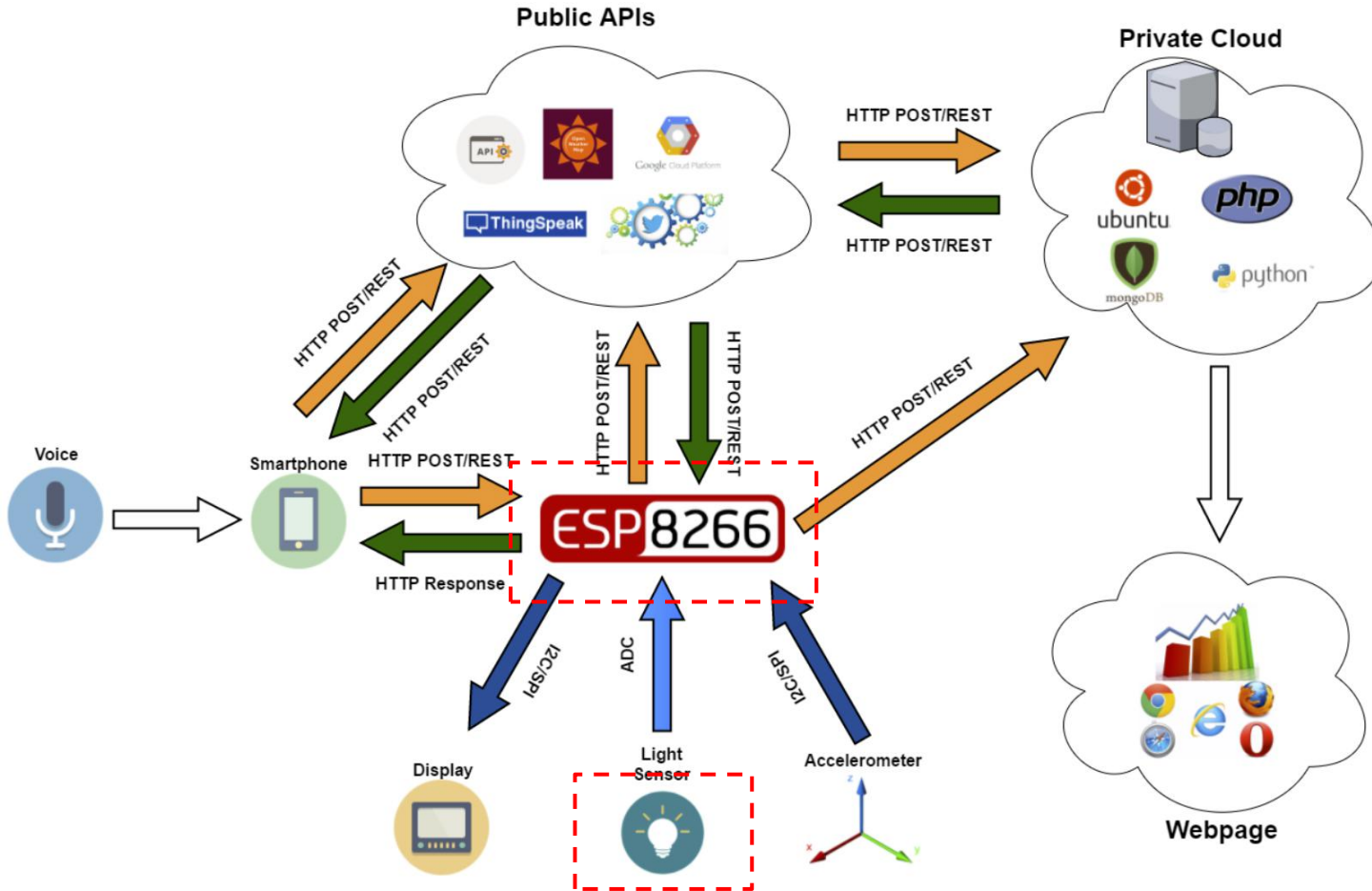
IoT - Intelligent and Connected Systems

EECS E4764 Fall' 22

Lab 2

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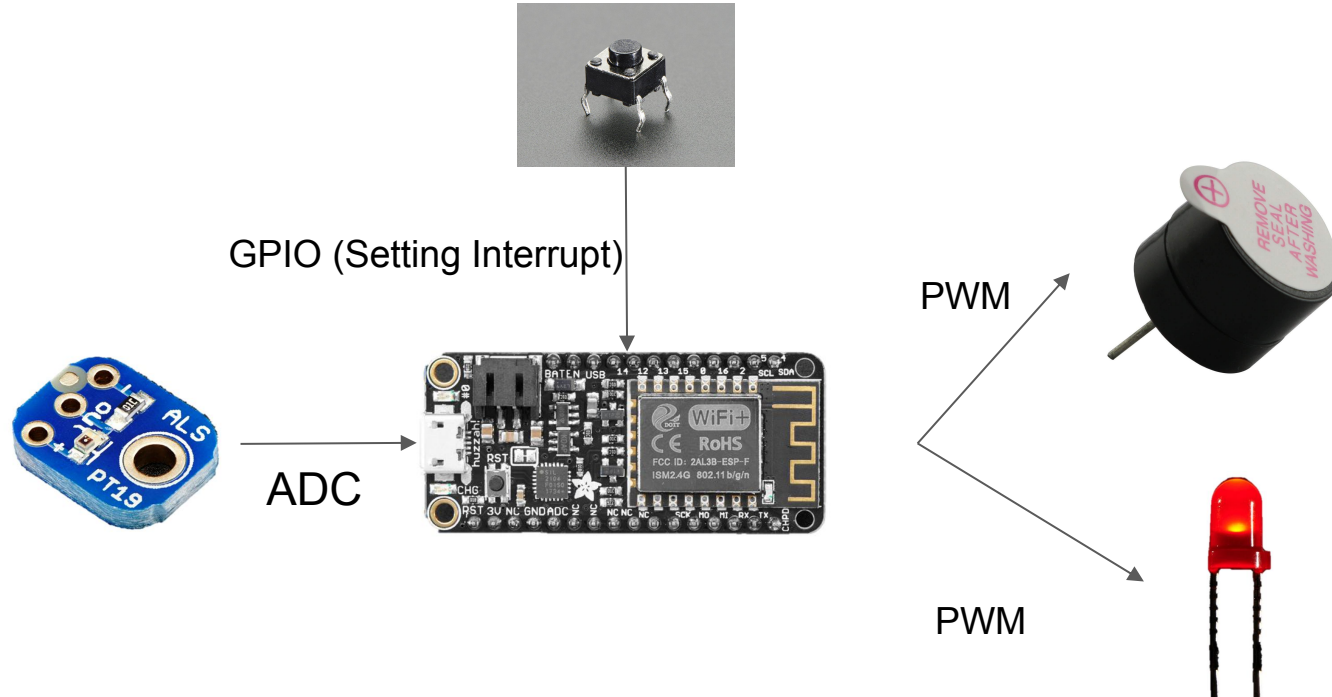
Lab Overview



Lab 2:

- Control peripheral outputs
- Read values from sensor data
- Vary peripheral outputs based on sensor inputs

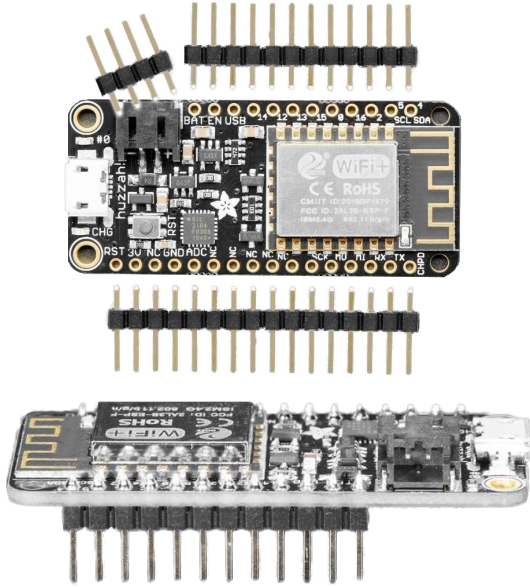
Lab 2 Overview



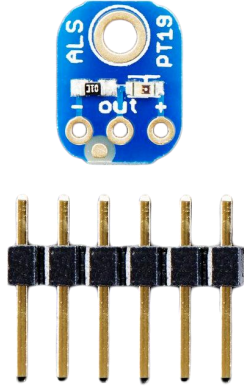
- <https://docs.micropython.org/en/latest/esp8266/quickref.html#pwm-pulse-width-modulation>
- <https://docs.micropython.org/en/latest/esp8266/quickref.html#adc-analog-to-digital-conversion>
- <http://docs.micropython.org/en/v1.9.3/esp8266/esp8266/tutorial/pins.html>
- <https://learn.adafruit.com/adafruit-feather-huzzah-esp8266/pinouts>

Soldering

Lab 2 (this week)



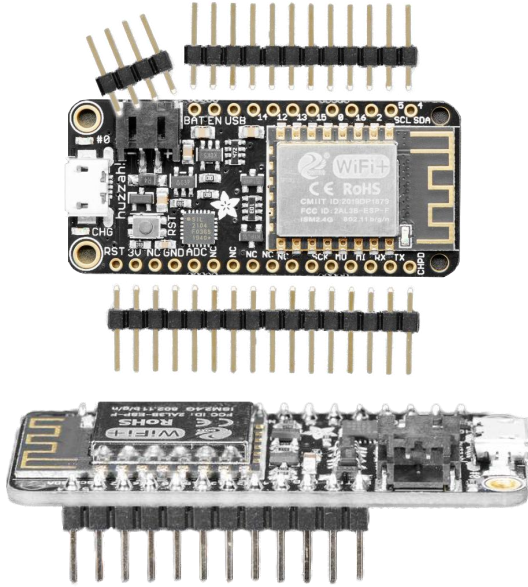
Feather HUZZAH



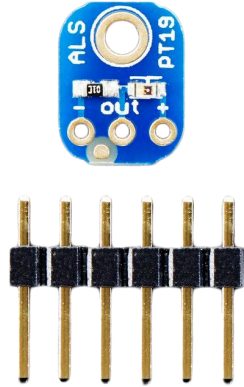
Light Sensor

Soldering

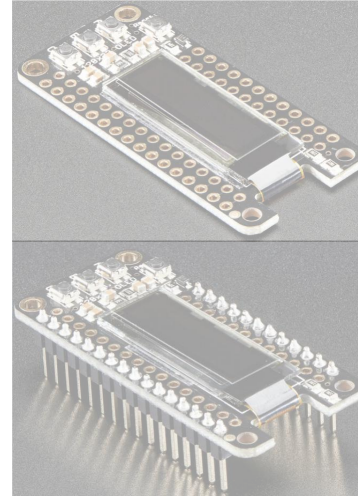
Lab 2 (this week)



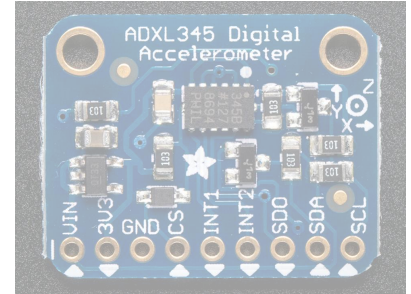
Feather HUZZAH



Light Sensor



OLED



Accelerometer

Lab 3 (next week)

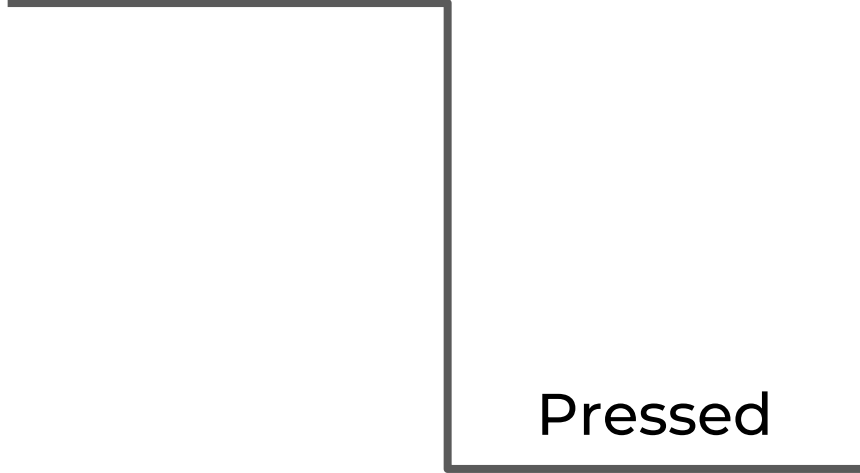
Pushbutton

Released

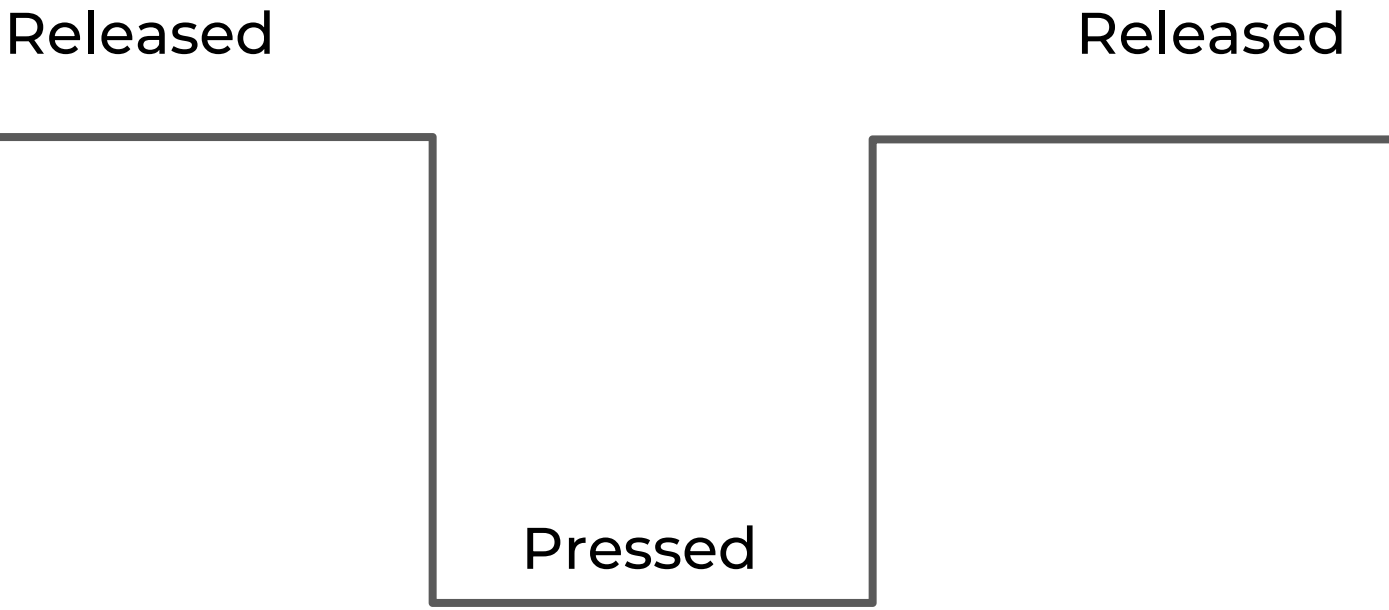
Pushbutton

Released

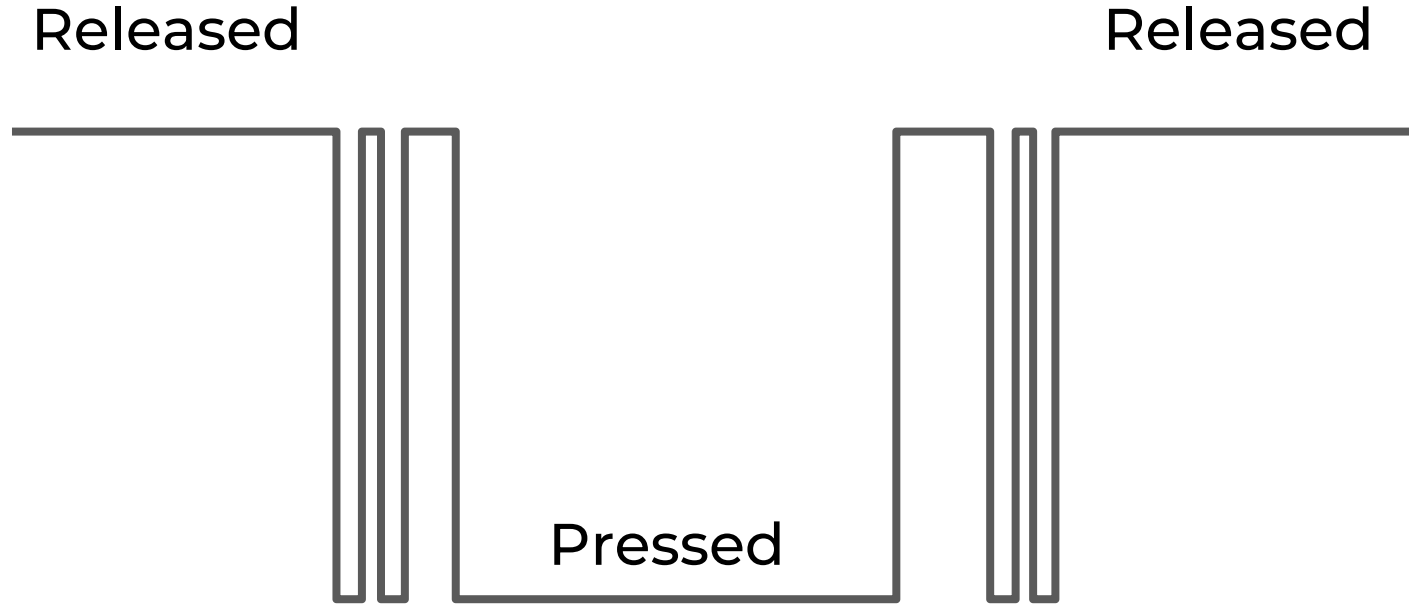
Pressed



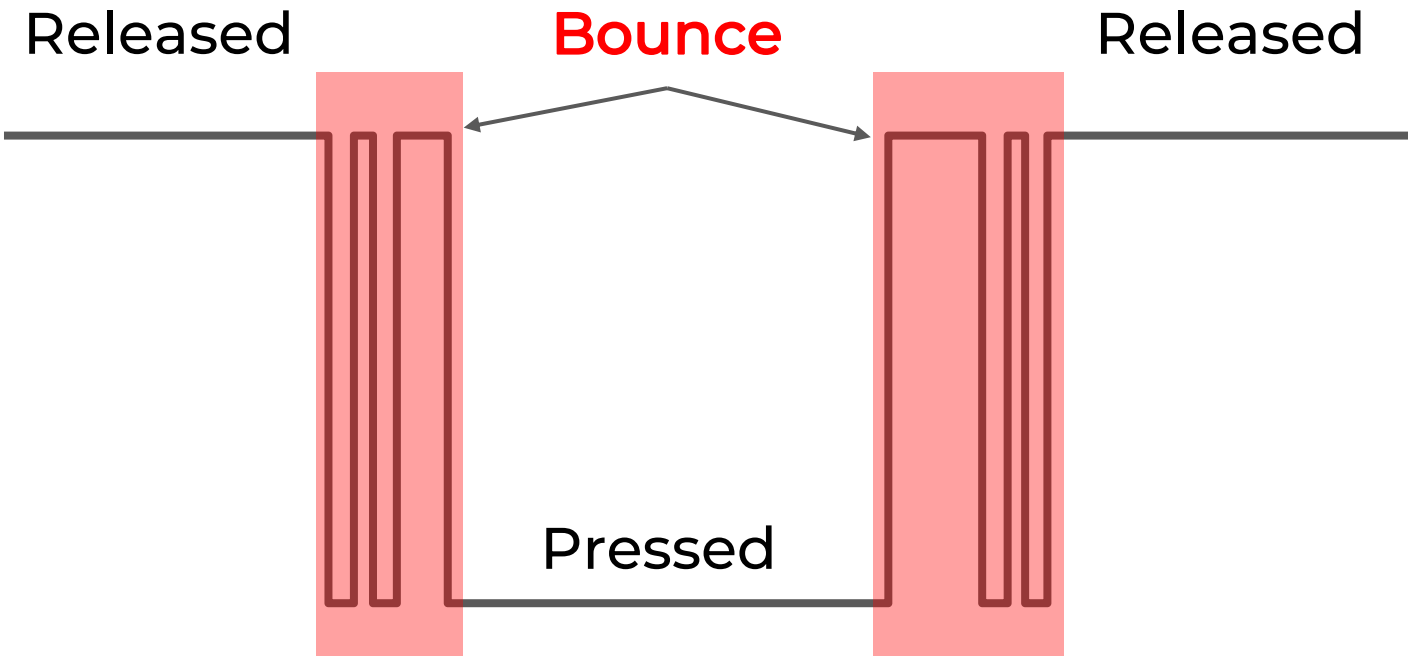
Pushbutton



It's not as easy as it seems



Debounceing



Lab 2 Checkpoints

1. Change the brightness of an LED and the pitch of a piezo in accordance to readings from the light sensor.
2. Debounce a button press. Use interrupt handler to deal with the button press.
3. Combine these two parts. Use button to control system on and off. The system is activated when the button is pressed and deactivated when released. While activated, the LED and piezo changes brightness and pitch depending on the light sensor readings.

Soldering Queue Sign-up Link

https://docs.google.com/spreadsheets/d/1kVPNzGXlpuskFbRAMIUquU6z6qsB7F9cm7_-HrYTWul/edit?usp=sharing

(if you can not go to the sheet, please check the address you enter, there might be some garbage characters like "%20" when copy and paste)